Teaching Guidelines for

# Logic Building & Problem Solving

*(Optional Preparatory Module)*

PGCP-AC February 2026

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**Duration:** 20 hours online sessions + problem-solving as self-study

**Objective:** This preparatory module is conducted online before the actual PG-DAC course commencement for students to learn to think logically and how to solve problems. A number of problems will be given to the students to solve them logically.

**Prerequisites:** Knowledge of computer fundamentals.

**Evaluation:** No separate evaluations. Daily assignment problems are to be solved and submitted.

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*Note: Each day comprises 3 hours of online lecture sessions followed by solving various problems using the topics learned by the students.*

**Day 1: Building Logic and Solving Problems**
**Lecture:**
Logical thinking
Problem-solving process
**Assignments:**
Practice logic building on given problems

**Day 2: Decision making**
**Lecture:**
If-else, nested if-else, switch case
**Assignments:**
Decision-making problems using if-else, nested if-else, switch case

**Day 3: Control Statements**
**Lecture:**
Loops
**Assignments:**
Conditional problems using loops

**Day 4: Modular Programming**
**Lecture:**
Functions
**Assignments:**
Problems using functions

**Day 5: Arrays**
**Lecture:**
Arrays
**Assignments:**
Problems using arrays

**Day 6: Strings**
**Lecture:**
Strings
**Assignments:**
Problems using strings